

Register Number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

22



Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Computer Science and Engineering

SAT

Regulations – R2021

Degree & Branch	BE-CSE				Semester	5
Subject Code & Name	UCS2503 SOFTWARE ENGINEERING					
Academic Year	2023-2024 ODD	Batch	2021-2025	Date	15.11.23	AN
Time: 01:30 - 03:00p.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1:	Choose an appropriate process model and estimate project cost and effort required by applying software engineering principles(K5)
CO2:	Identify and analyse the requirements and construct their models(K5)
CO3:	Apply systematic procedure for software design(K3)
CO4:	Compare and contrast the various testing and maintenance activities (K2)
CO5:	Make use of agile development and Devops(K3)
CO6:	Identify unethical issues and apply ethical practices for a given casestudy(K3)

Part – A (4 × 2 = 8Marks)

		KL	CO	PI
1.	List a few best practices followed in software organization for successful software development.	K1	CO1	1.3.1
2.	Which phase in the SDLC is most important to deliver the software product?	K1	CO1	2.1.2
3.	List the main reasons to use Test Automation tools.	K1	CO4	3.4.1 13.3.2
4.	Explain the importance of regression testing in software development.	K2	CO4	3.4.1 13.3.2

Part – B (3×6 = 18 Marks)

		KL	CO	PI
5.	Differentiate between waterfall model and spiral model.	K2	CO1	1.3.1 1.4.1
6.	Identify the risk in developing embedded software which remotely monitors a car and how the project engineer will ensure risk reduction during software development?	K3	CO1	1.3.1 1.4.1
7.	Build the security test cases for the test scenarios given below. - A password should be in encrypted format. - Application or System should not allow invalid users. - For financial sites, the Browser back button should not work.	K3	CO4	2.2.1 13.3.2 14.1.2

Part – C (2 × 12 = 24 Marks)
(Any subdivisions should be either 7+5 or 8+4)

(Any subdivisions should be either 7+5 or 8+4)

		KL	CO	PI																											
8.	a) Use the Table 1 and estimate the function point to develop the software based on the following data. (7) - A target product has 15 simple inputs, 3 average inputs, 60 average outputs, 6 simple outputs, 18 simple inquiries, 10 average master files, and 7 complex interfaces. - The total degree of influence for the product is 46. b) Given a 34 FP, calculate the effort in person month required to complete the project. Note: organizational effort = 12 FP person month. Table 1: Function point evaluation <table><tr><th rowspan="2">Component</th><th colspan="3">Level of Complexity</th></tr><tr><th>Simple</th><th>Average</th><th>Complex</th></tr><tr><td>External Inputs</td><td>3</td><td>4</td><td>6</td></tr><tr><td>External outputs</td><td>4</td><td>5</td><td>7</td></tr><tr><td>External Inquiries</td><td>3</td><td>4</td><td>6</td></tr><tr><td>Internal logical files</td><td>7</td><td>10</td><td>15</td></tr><tr><td>External interface files</td><td>5</td><td>7</td><td>10</td></tr></table>	Component	Level of Complexity			Simple	Average	Complex	External Inputs	3	4	6	External outputs	4	5	7	External Inquiries	3	4	6	Internal logical files	7	10	15	External interface files	5	7	10	K3	CO1	1.3.1 2.1.2
Component	Level of Complexity																														
	Simple	Average	Complex																												
External Inputs	3	4	6																												
External outputs	4	5	7																												
External Inquiries	3	4	6																												
Internal logical files	7	10	15																												
External interface files	5	7	10																												

(Or)

9	a) What are the different types of scheduling techniques in software engineering? Which technique is the best one? Why? (5)	K3	CO1	1.3.1																											
	b) Apply the scheduling technique and identify the critical path activities. (7)			2.1.2																											
	<table><tr><th>Activity</th><th>Immediate Predecessor(s)</th><th>Time (Days)</th></tr><tr><td>A</td><td>-</td><td>3</td></tr><tr><td>B</td><td>A</td><td>4</td></tr><tr><td>C</td><td>A</td><td>6</td></tr><tr><td>D</td><td>B</td><td>6</td></tr><tr><td>E</td><td>B</td><td>4</td></tr><tr><td>F</td><td>C</td><td>4</td></tr><tr><td>G</td><td>D</td><td>6</td></tr><tr><td>H</td><td>E, F</td><td>8</td></tr></table>	Activity	Immediate Predecessor(s)	Time (Days)	A	-	3	B	A	4	C	A	6	D	B	6	E	B	4	F	C	4	G	D	6	H	E, F	8			
Activity	Immediate Predecessor(s)	Time (Days)																													
A	-	3																													
B	A	4																													
C	A	6																													
D	B	6																													
E	B	4																													
F	C	4																													
G	D	6																													
H	E, F	8																													

10.	a) Explain any two of the basis path testing methods to identify the number of independent paths. (7 Marks) b) Write a program to print Fibonacci series and generate the test cases to test your code using basis path testing (5 Marks)	K2	CO4	2.2.1 13.3.2 14.1.2
-----	---	----	-----	---------------------------

(Or)

11.	a) Show how to apply equivalence class partitioning testing to a code? Is equivalence partitioning better than boundary value analysis? (7) b) Why is integration testing more important than unit testing? (5)	K2	CO4	2.2.1 13.3.2 14.1.2
-----	---	----	-----	---------------------------